COMPUTING 2 - PARALLEL COMPUTING

Chapter 3 Mock Test

GPU Architecture and SIMT Execution

Suggested time	Total points	Coverage
50 minutes	60	Grid/block/thread hierarchy, warps and wavefronts, divergence, latency hiding, memory spaces

NAME		STUDENT ID	
DATE		SCORE	/ 60

Instructions
- Answer every question and show essential reasoning for partial credit.
- For true/false items, an unsupported letter receives no credit.
- Unless stated otherwise, assume NVIDIA warp size 32 and ideal arithmetic.
- Keep the answer section closed until the suggested time has elapsed.

Questions

3.1 True or false with justification — 8 points

(a) GPU cores generally devote more area to throughput than to aggressive control logic.

(b) Software chooses the hardware warp width for each kernel launch.

(c) Threads in one CUDA block may synchronize with __syncthreads().

(d) __syncthreads() is a grid-wide barrier.

(e) Divergence is absent when every active lane follows the same path.

(f) Shared memory is allocated per block.

(g) Historical AMD GCN commonly uses 64-work-item wavefronts.

(h) Arm Valhall performs dynamic instruction scheduling in hardware.

WORK AREA

3.2 Launch accounting — 8 points

A kernel launches 1000 blocks of 512 threads on an NVIDIA-style machine with 32-thread warps.

Find (a) logical threads, (b) warps per block, (c) logical warps in the launch, and (d) the number of lifetime instances of one scalar __shared__ variable.

WORK AREA

3.3 Resource-limited occupancy — 12 points

One SM supports 2048 resident threads, 32 resident blocks, 65,536 registers, and 64 KiB shared memory. A block has 256 threads, uses 64 registers/thread and 20 KiB shared memory.

How many blocks can reside on the SM? What is thread occupancy? Recompute if shared memory falls to 16 KiB with all else unchanged.

WORK AREA

3.4 Divergence cost — 8 points

In one warp, 16 lanes take path A (20 cycles) and 16 take path B (8 cycles). Assume predicated serial execution of both paths and ignore reconvergence overhead.

Find elapsed path time and lane-utilization efficiency. Compare with a warp in which all lanes take A.

WORK AREA

3.5 Coalescing from address patterns — 8 points

A 32-thread warp reads 4-byte floats from a 128-byte aligned base. Compare A[t] with A[2t]. Assume 128-byte transaction segments for this exercise.

How many segments are touched and what useful-byte fraction is obtained?

WORK AREA

3.6 Architecture comparison — 8 points

Briefly compare NVIDIA CUDA, historical AMD GCN, and Arm Valhall using the terms warp/wavefront, workgroup/block residency, and scheduling.

WORK AREA

3.7 Choose a GPU memory space — 8 points

Choose registers, shared memory, constant memory, texture/read-only cache, global memory, or Unified Memory for each case: (a) one thread's accumulator, (b) a block-reused tile, (c) a small read-only scalar table read uniformly, (d) a spatially accessed 2D image, (e) a large device array, (f) one convenient CPU/GPU pointer.

WORK AREA

ANSWER SECTION

Chapter 3: Answer Key & Worked Solutions

Grade the method as well as the final result. Numerical answers show enough intermediate work to identify where an error occurred.

Quick Answer Key

Q	Topic	Short answer
3.1	True or false with justification	T, F, T, F, T, T, T, T
3.2	Launch accounting	512,000 threads; 16 warps/block; 16,000 warps; 1,000 shared instances
3.3	Resource-limited occupancy	3 blocks, 37.5%; then 4 blocks, 50%
3.4	Divergence cost	28 cycles and 50% lane efficiency; all-A takes 20 cycles at 100% lane use
3.5	Coalescing from address patterns	A[t]: 1 segment, 100%; A[2t]: 2 segments, 50%
3.6	Architecture comparison	NVIDIA warp 32; GCN wavefront commonly 64; Valhall uses hardware dynamic scheduling in the lecture model
3.7	Choose a GPU memory space	registers; shared; constant; texture/read-only; global; Unified Memory

Worked Solutions

3.1 True or false with justification — 8 points

Answer. (a) T, (b) F, (c) T, (d) F, (e) T, (f) T, (g) T, (h) T for the lecture's Valhall characterization.

- Warp/wavefront width is an architectural property.
- A block-local barrier cannot directly synchronize blocks.
- The lecture describes Valhall as using hardware dynamic instruction scheduling and superscalar issue.

Common trap. Architecture-specific values should follow the generation and assumptions stated in the question.

3.2 Launch accounting — 8 points

Answer. Count logical work independently of how many blocks are resident simultaneously.

- Threads=1000x512=512,000.
- Warps/block=512/32=16.
- Logical warps=1000x16=16,000.
- Shared storage is per block, so 1000 versions are created over the kernel lifetime.

Common trap. Do not multiply shared-variable versions by thread count. Threads in a block share one instance.

3.3 Resource-limited occupancy — 12 points

Answer. Take the minimum block limit induced by every resource.

- Thread limit=floor(2048/256)=8 blocks.
- Register limit=floor(65536/(256x64))=4 blocks.
- Shared-memory limit=floor(64/20)=3 blocks; architectural block limit=32.
- Resident blocks=min(8,4,3,32)=3; occupancy=3x256/2048=37.5%.
- At 16 KiB/block, the shared limit becomes 4. Registers also limit to 4, so occupancy=4x256/2048=50%.

$$\text{resident blocks} = \min(\text{thread}, \text{register}, \text{shared-memory}, \text{block-count limits})$$

3.4 Divergence cost — 8 points

Answer. When both paths occur, the warp executes A and B serially with nonparticipating lanes masked.

- Elapsed time=20+8=28 cycles.
- Useful lane-cycles=16x20+16x8=448.
- Available lane-cycles=32x28=896, so efficiency=448/896=50%.
- All lanes taking A use 32x20 useful lane-cycles in 20 cycles: 100%.

Common trap. The hardware preserves correctness automatically; it does not remove the performance cost.

3.5 Coalescing from address patterns — 8 points

Answer. Inspect the set of addresses generated by the whole warp, not whether each thread accesses adjacent elements over time.

- A[t] spans byte offsets 0 through 124: one 128-byte segment, 128 useful bytes.
- A[2t] spans offsets 0 through 248 and touches two segments. It still requests 128 useful bytes from 256 transferred bytes: 50%.

Common trap. Real transaction behavior varies by architecture and cache level; this answer follows the explicit 128-byte-segment assumption.

3.6 Architecture comparison — 8 points

Answer. All three execute groups of lanes, but terminology and scheduling details differ.

Architecture	Lane group	Residency/scheduling point
NVIDIA CUDA	Warp, commonly 32	Blocks reside on an SM; warp scheduler issues ready warps
AMD GCN	Wavefront, commonly 64	All wavefronts of a workgroup reside on one CU; CU holds many wavefronts
Arm Valhall	Warp-like grouped execution	Lecture model emphasizes hardware dynamic instruction scheduling and superscalar issue

3.7 Choose a GPU memory space — 8 points

Answer. Match scope, access pattern, and ownership rather than choosing the smallest memory automatically.

- (a) Registers, (b) shared memory, (c) constant memory, (d) texture/read-only path, (e) global memory, (f) Unified Memory via cudaMallocManaged.
- Unified Memory improves programmability but migration and prefetch behavior can affect performance.